Exercises C: Coursework

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1. Consider the SDE

$$dX(t) = (\mu - X(t))\,dt + \sigma\,dW(t), \quad X(0) = x_0.$$

where μ, σ are positive constants and $W(t)$ standard Brownian motion in 1 dimension.

   (a) Solve this equation using an integrating factor (variation of constants formula).

   (b) Calculate the mean and the variance when the initial condition x is deterministic.

   (c) Write down the generator $\mathbb{L}$ and the corresponding forward Fokker-Planck equation

   (d) Find the steady-state solution of the FP equation and show that the resulting mean and variance are consistent with the $t \to \infty$ limit of part (b).

2. Let $W(t)$ be a standard one dimensional Brownian motion.

   (a) Obtain the solution of the SDE

$$dX(t) = \left(\frac{1}{4} - X(t)\right)\,dt + \sqrt{X(t)}\,dW(t), \quad X(0) = x \geq 0.$$

   Hint: Perform the change of variables $Y(t) = 2\sqrt{X(t)}$ using Itô's formula and show that $Y(t)$ satisfies an OU process.

   (b) Obtain the solution of the Stratonovich stochastic differential equation

$$dX(t) = f(X(t)) \circ dW(t) \quad X(0) = x_0,$$

   where f is a positive differentiable function. Hint: convert to an Itô SDE and then perform the change of variables

$$Y(t) = F(X(t)) \equiv \int^{X(t)} \frac{1}{f(y)}\,dy.$$

   (c) Use Part (b) to show that solution of the Itô stochastic differential equation

$$dX(t) = -\frac{1}{2}X(t)\,dt + \sqrt{1 - X(t)^2}\,dW(t), \quad X(0) = x_0 \in [-1, 1].$$

   is

$$X(t) = \sin(W(t) + \arcsin(x_0)).$$

3. Consider the system of Itô stochastic differential equations

$$\begin{aligned} dX(t) &= Y(t)\,dt, \quad X(0) = x_0, && \text{(1a)} \\ dY(t) &= -X(t)\,dt + Z(t)\,dt, \quad Y(0) = y_0, && \text{(1b)} \\ dZ(t) &= -Z(t)\,dt - Y(t)\,dt + \sqrt{2}\,dW(t), \quad Z(0) \sim \mathcal{N}(0,1). && \text{(1c)} \end{aligned}$$

(a) Write down the generator $\mathbb{L}$, its L^2-adjoint $\mathbb{L}^*$ and the forward (Fokker-Planck) Kolmogorov equation for the process $\mathbf{X}(t) = (X(t),\, Y(t),\, Z(t))$.

(b) Show that (1) can be written in the form

$$\frac{dX(t)}{dt} = Y(t), \quad \frac{dY(t)}{dt} = -X(t) - \int_0^t e^{-(t-s)} Y(s)\, ds + F(t), \tag{2}$$

where $F(t)$ is a mean zero stationary Gaussian process with autocorrelation function

$$\mathbb{E}(F(t)F(s)) = e^{-|t-s|}.$$

(c) Is the process $(X(t),\, Y(t))$ defined in (2) a Markov process?

4. Consider the multivariate Ornstein-Uhlenbeck process.

$$dX_i = \sum_{j=1}^{d} A_{ij} X_j(t) dt + \sum_{j=1}^{d} B_{ij} dW_j(t).$$

(a) Show that the solution in vector form is given by

$$\mathbf{X}(t) = \mathrm{e}^{\mathbf{A}t}\bar{\mathbf{x}} + \int_0^t \mathrm{e}^{\mathbf{A}(t-t')} \mathbf{B} d\mathbf{W}(t').$$

(b) Introduce the correlation function $\mathbf{C}(t,s) = \langle \mathbf{X}(t), \mathbf{X}^T(s) \rangle$ with components

$$C_{ij}(t,s) = \langle X_i(t), X_j(s) \rangle = \langle [X_i(t) - \langle X_i(t) \rangle][X_j(s) - \langle X_j(s) \rangle] \rangle.$$

Using part (a), show that

$$\mathbf{C}(t,s) = \int_0^{\min(t,s)} \mathrm{e}^{\mathbf{A}(t-t')} \mathbf{B}\mathbf{B}^T \mathrm{e}^{\mathbf{A}^T(s-t')} dt'.$$

(c) Introduce the covariance matrix $\Sigma(t) = \mathbf{C}(t,t)$ with components

$$\Sigma_{ij}(t) = \langle [X_i(t) - \langle X_i(t) \rangle][X_j(t) - \langle X_j(t) \rangle] \rangle.$$

Derive the matrix equation

$$\frac{d\Sigma(t)}{dt} = \mathbf{A}\Sigma(t) + \Sigma(t)\mathbf{A}^T + \mathbf{B}\mathbf{B}^T.$$

Hence, show that if $\mathbf{A}$ has distinct eigenvalues with negative real part, then $\Sigma(t) \to \Sigma_0$ where Σ_0 is the stationary covariance matrix satisfying

$$\mathbf{A}\Sigma_0 + \Sigma_0 \mathbf{A}^T + \mathbf{B}\mathbf{B}^T = 0.$$

(d) By formally setting $dW_i(t) = \eta_i(t) dt$ with

$$\langle \xi_i(t) \rangle = 0, \quad \langle \xi_i(t) \xi_j(t') \rangle = \delta_{i,j} \delta(t - t'),$$

show that the spectrum of the i-th component $X_i(t)$ is

$$S_i(\omega) = \sum_{j=1}^{d} \sum_{j'=1}^{d} \Phi_{ij}^{-1}(\omega)\Phi_{ij'}^{-1}(-\omega)D_{jj'},$$

where $D_{ij} = \sum_k B_{ik}B_{kj}^{\top}$ and

$$\Phi_{ij}(\omega) = -i\omega\delta_{i,j} - A_{ij}.$$

5. Consider the Itô SDE

$$dX(t) = (-X(t)^3 + X(t))\,dt + \sqrt{2}\,dW(t), \quad X(0) = x,$$

where $x \sim \rho_0(x)$ with finite moments of all orders.

(a) Write down the generator and the backward and forward Kolmogorov equations for the process $X(t)$.

(b) Use the forward Kolmogorov equation to obtain a system of equations for the moments of $X(t)$, $m_n(t) := \mathbb{E}[X(t)^n]$, $n = 0, 1, 2 \ldots$.

(c) Show that the solution of the stationary FP equation $\mathbb{L}^* p(x) = 0$ is $p^*(x) = Z^{-1}\mathrm{e}^{-V(x)}$ and determine $V(x)$.

(d) Suppose that there exists another stationary solution of the form $h(x)p^*(x)$. Show that $h(x)$ satisfies the backward equation $\mathbb{L}h(x) = 0$. Finally, multiplying the latter by $h(x)p_*(x)$ and integrating with respect to x show that

$$\int_{\mathbb{R}} |h'|^2 p_*\,dx = 0,$$

and hence deduce that $X(t)$ is ergodic.